

1.Data Ingestion Process

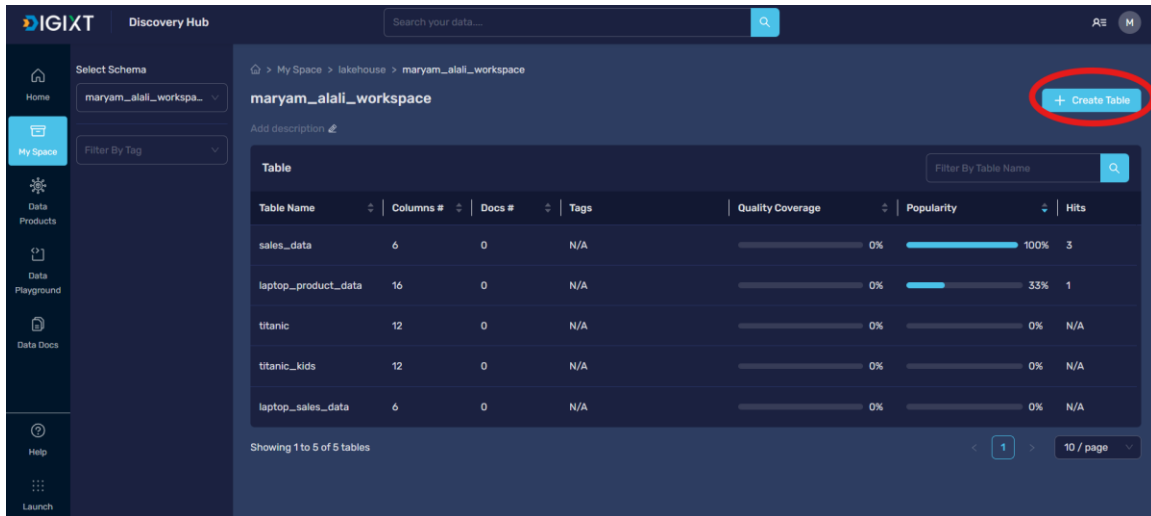


Figure 1: Ingest the data into DigiXT

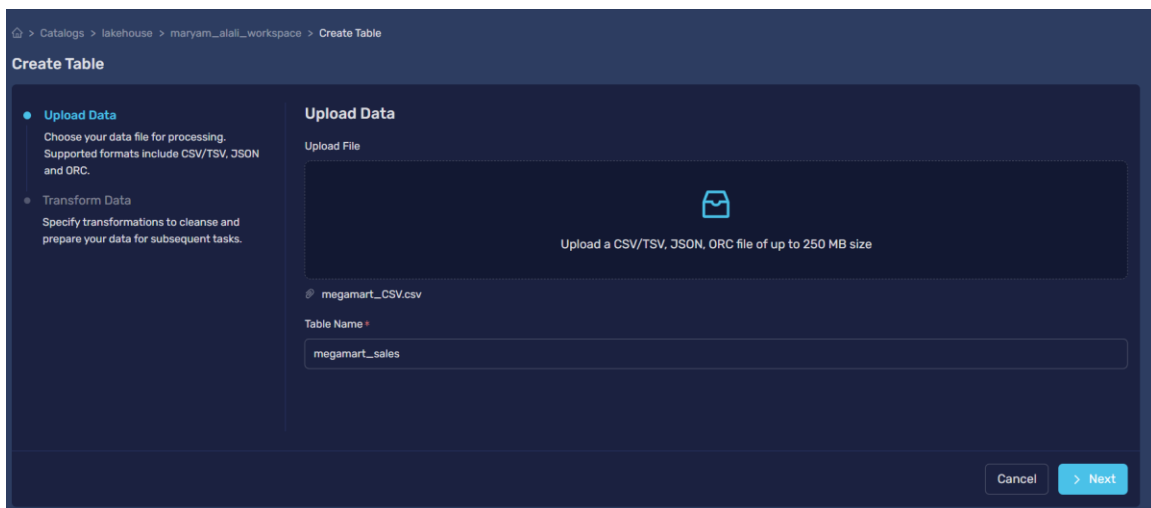


Figure 2: Upload data

The dataset was successfully loaded into the DigiXT platform. The process confirmed that all records were transferred correctly, with numerical columns like sales and profit

2 Initial Test Suite Results (Before Cleaning)

We ran a series of automated tests to check the data for Completeness, Consistency, Validity, and Accuracy

My Space > lakehouse > manyam_atl_workspace > megarmlt_sales > Create Rule

Create Rule

- Choose/Create Test Suite**
Choose a test suit or create a test suite to add rules.
- Manage Table Filters**
Utilize filters to manage the data included or excluded from analysis or reporting, governing precision and relevance.
- Define Rules**
Data quality rules play a critical role in data governance and management, ensuring adherence to predefined standards.

Create Test Suite

Choose/Create Test Suite: New

Test Suite Name: clean_min_quality

Description: This test suite contains the quality rules on the megarmlt_sales

Execution Schedule: ☒ Hourly ☐ Daily ☐ Weekly ☐ Monthly ☐ Yearly

Select Time: 01

Cancel Next

Figure 3: Create test suite

Define Rules

Type: Column Category: Accuracy Rule Name: Max Value Check

Column: quantity Min Value: 0 Max Value: 20 Strict Min: Yes

Strict Max: Yes Parse Strings As Datetimes: No Output Strftime Format:

+ Add

Name	Type	Category	Arguments	Column	Actions
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Figure 4: Rule 1 Max value check

Define Rules

Type: Column Category: Accuracy Rule Name: Min Value Check

Column: quantity Min Value: 0 Max Value: 20 Strict Min: Yes

Strict Max: Yes Parse Strings As Datetimes: No Output Strftime Format:

+ Add

Name	Type	Category	Arguments	Column	Actions
Max Value Check	Column	Accuracy	max_value: 20 strict_min: true + 2	quantity	

< 1 >

Figure 5: Rule 2 Min value check

Define Rules

Type

All

Category

All

Rule Name

Row Count Match

Value

6357

+ Add

Name	Type	Category	Arguments	Column	Actions
Row Count Range	Table	Completeness	min_value: 6000 max_value: 10000 +1	N/A	
Max Value Check	Column	Accuracy	max_value: 20 strict_min: true +2	quantity	
Min Value Check	Column	Accuracy	max_value: 20 strict_min: true +2	quantity	

< 1 >

Figure 6: Rule 3 Row count match

Define Rules

Type

All

Category

All

Rule Name

Float Type Check

Column

sales

Mostly

+ Add

Name	Type	Category	Arguments	Column	Actions
Row Count Match	Table	Completeness	value: 6357 catch_exceptions: true	N/A	
Row Count Range	Table	Completeness	min_value: 6000 max_value: 10000 +1	N/A	
Max Value Check	Column	Accuracy	max_value: 20 strict_min: true +2	quantity	
Min Value Check	Column	Accuracy	max_value: 20 strict_min: true +2	quantity	

< 1 >

Figure 7: Rule 4 Float type check

Define Rules

Type

All

Category

All

Rule Name

String Type Check

Column

region

Mostly

+ Add

Name	Type	Category	Arguments	Column	Actions
Float Type Check	Column	Validity	type_: "float" catch_exceptions: true	sales	
Row Count Match	Table	Completeness	value: 6357 catch_exceptions: true	N/A	
Row Count Range	Table	Completeness	min_value: 6000 max_value: 10000 +1	N/A	
Max Value Check	Column	Accuracy	max_value: 20 strict_min: true +2	quantity	
Min Value Check	Column	Accuracy	max_value: 20 strict_min: true +2	quantity	

< 1 >

Figure 8: Rule 5 String type check

Define Rules

Type

All

Category

All

Rule Name

Unique Values Match All in List

Column

region

Value Set

Central

South

East

West

Parse Strings As Datetimes

No

+ Add

Name	Type	Category	Arguments	Column	Actions
String Type Check	Column	Validity	type: "str" catch_exceptions: true	region	
Float Type Check	Column	Validity	type: "float" catch_exceptions: true	sales	
Row Count Match	Table	Completeness	value: 6357 catch_exceptions: true	N/A	
Row Count Range	Table	Completeness	min_value: 6000 max_value: 10000 +1	N/A	
Max Value Check	Column	Accuracy	max_value: 20 strict_min: true +2	quantity	
Min Value Check	Column	Accuracy	max_value: 20 strict_min: true +2	quantity	

Figure 9: Rule 6 Unique Values Match All in list

Define Rules

Type

All

Category

All

Rule Name

Column Count Range

Min Value

15

Max Value

20

+ Add

Name	Type	Category	Arguments	Column	Actions
Unique Values Match All in List	Column	Uniqueness	value_set: Central South East West catch_exceptions: true	region	
String Type Check	Column	Validity	type: "str" catch_exceptions: true	region	
Float Type Check	Column	Validity	type: "float" catch_exceptions: true	sales	
Row Count Match	Table	Completeness	value: 6357 catch_exceptions: true	N/A	
Row Count Range	Table	Completeness	min_value: 6000 max_value: 10000 +1	N/A	
Max Value Check	Column	Accuracy	max_value: 20 strict_min: true +2	quantity	

Figure 10: Rule 7 Column Count Range

Define Rules

Type

All

Category

All

Rule Name

Not Null Value Check

Column

order_id

Mostly

+ Add

Name	Type	Category	Arguments	Column	Actions
Column Count Range	Table	Completeness	min_value: 15 max_value: 20 +1	N/A	
Unique Values Match All in List	Column	Uniqueness	value_set: Central.South.East.West catch_exceptions: true	region	
String Type Check	Column	Validity	type_: "str" catch_exceptions: true	region	
Float Type Check	Column	Validity	type_: "float" catch_exceptions: true	sales	
Row Count Match	Table	Completeness	value: 657 catch_exceptions: true	N/A	
Row Count Range	Table	Completeness	min_value: 6000 max_value: 10000 +1	N/A	

Figure 11: Rule 8 Not Null Value Check

Define Rules

Type

All

Category

All

Rule Name

Unique Table Rows Check

Column List

order_id x +2

Ignore Row If

All values are missing

+ Add

Name	Type	Category	Arguments	Column	Actions
Not Null Value Check	Column	Completeness	catch_exceptions: true	order_id	
Column Count Range	Table	Completeness	min_value: 15 max_value: 20 +1	N/A	
Unique Values Match All in List	Column	Uniqueness	value_set: Central.South.East.West catch_exceptions: true	region	
String Type Check	Column	Validity	type_: "str" catch_exceptions: true	region	
Float Type Check	Column	Validity	type_: "float" catch_exceptions: true	sales	
Row Count Match	Table	Completeness	value: 657 catch_exceptions: true	N/A	

Figure 12: Rule 9 Unique Table Rows Check

My Space

lakehouse

maryam_atall_workspace

megamart_sales

megamart_sales

78% Quality %

7/9 Successful Rules

0 Skipped Rules

0 Queued Rules

Summary

Rules

Settings

Run

+ Rule

Show: 1 week

Filter

All Categories

Uniqueness

Completeness

Validity

Accuracy

Unique Table Rows Check - Last evaluated on 11/20/2025 at 7:01:02 PM (Asia/Dubai)

Uniqueness

Passed

Megamart Initial Quality Check

Not Null Value Check - Last evaluated on 11/20/2025 at 7:01:02 PM (Asia/Dubai)

Completeness

Passed

Megamart Initial Quality Check

Column Count Range - Last evaluated on 11/20/2025 at 7:01:02 PM (Asia/Dubai)

Completeness

Passed

Megamart Initial Quality Check

Unique Values Match All in List - Last evaluated on 11/20/2025 at 7:01:02 PM (Asia/Dubai)

Uniqueness

Passed

Megamart Initial Quality Check

String Type Check - Last evaluated on 11/20/2025 at 7:01:02 PM (Asia/Dubai)

Validity

Passed

Megamart Initial Quality Check

Float Type Check - Last evaluated on 11/20/2025 at 7:01:02 PM (Asia/Dubai)

Validity

Passed

Megamart Initial Quality Check

Row Count Match - Last evaluated on 11/20/2025 at 7:01:02 PM (Asia/Dubai)

Completeness

Failed

Megamart Initial Quality Check

Min Value Check - Last evaluated on 11/20/2025 at 7:01:02 PM (Asia/Dubai)

Accuracy

Failed

Megamart Initial Quality Check

Max Value Check - Last evaluated on 11/20/2025 at 7:01:02 PM (Asia/Dubai)

Accuracy

Passed

Megamart Initial Quality Check

Figure 13: Data quality check before cleaning

3. Data Preprocessing Strategy

3.1. Preprocessing Objective and Methodology

By comparing the raw data file (megamart_CSV.csv) with the cleaned data file (cleaned_megamart.csv), we identified and documented the exact cleaning and preprocessing steps that were applied.

3.2. Identified Cleaning Steps and Rationale

Our comparison showed that the main data quality issues found in Task 4 were solved in the cleaned dataset, but by deleting records instead of filling in missing values. The table below documents the actions taken and the logic behind them:

Quality Issue Identified (in Raw Data)	Actions taken	Rationale & Advanced Technique
Missing Values in Key Columns (order_date, quantity, review)	Record Deletion	All records that had any missing value in these three columns were deleted.
Negative Quantity (9 records)	Record Deletion	The 9 records with negative quantities were deleted. They were likely part of the records that had missing values.
Extreme Outliers (profit)	No Action Taken (Preservation)	The outliers in the profit column were kept. Technique: Strategic Preservation, recognizing that these are not errors but real business events (like large deals or losses) that are important for a realistic analysis.

4. Re-conducting the Quality Check (Post-Curation)

To check if the applied changes were effective, we performed a final quality check on the cleaned_megamart.csv file.

Validation Check	Raw Data State	Clean Data State	Status
Total Records	9,994	6358	VALIDATED: 3,637 records were deleted. This solves the missing value problem.
Missing order_date	2,278	0	VALIDATED: All missing dates were deleted.
Missing quantity	1,257	0	VALIDATED: All missing quantities were deleted.
Negative quantity Count	9	0	VALIDATED: All negative values were deleted.

Conclusion: The change from the raw to the clean dataset significantly improved data quality in terms of completeness and validity.

The cleaned data was provided and carefully cleaned by the course instructor

The screenshot displays a data quality dashboard for the 'megamartt_sales' table. The top section shows a '100% Quality %' badge, '10/10 Successful Rules', '0 Skipped Rules', and '0 Queued Rules'. Below this, a 'Summary' tab is active, showing a list of rules categorized by 'All Categories', 'Uniqueness', 'Completeness', 'Validity', and 'Accuracy'. Each rule entry includes a name, a last evaluation timestamp, and a status indicator (e.g., 'Uniqueness', 'Completeness', 'Validity') with a 'Passed' label and a 'clean_data_quality' link. The rules listed are: Unique Table Rows Check, Not Null Value Check, Column Count Range, Unique Values Match All in List, String Type Check, Float Type Check, Row Count Match, Row Count Range, Max Value Check, and Min Value Check.

Figure 14: Data quality check after cleaning